

Executive Summary

Customer-support voice assistants often have to work with background noise, different microphones, telephone-quality audio, and different accents. These conditions can reduce speech recognition accuracy, and transcription mistakes can affect later tasks such as intent detection, retrieval, summarization, and response generation.

Scope of the Study

This project compares **Whisper Small, Faster-Whisper Small, and Wav2Vec2 Base 960h**. The comparison considers not only transcription accuracy but also inference speed, memory usage, deployment complexity, and performance under noisy conditions.

Experimental Approach

All three models are tested using the same open speech dataset under clean and controlled noisy conditions. This keeps the comparison consistent because each model processes the same audio and reference transcripts. The main evaluation metrics are **Word Error Rate (WER), inference latency, Real-Time Factor (RTF), and approximate memory usage**.

Model Selection

Whisper Small is used as the general-purpose baseline and follows an encoder-decoder Transformer architecture. **Faster-Whisper Small** uses the Whisper model with the CTranslate2 inference framework, making it useful for comparing inference efficiency. **Wav2Vec2 Base 960h** uses a different CTC-based approach and provides an established English ASR baseline.

Key Finding

Clean speech results alone may not show how a model will perform in a real customer-support environment. The study therefore compares both clean and noisy results and looks at how much the WER changes after noise is introduced.

Conclusion

The final model should be selected based on the overall balance of **WER, noise robustness, latency, memory usage, and deployment requirements**. Additional testing with representative, properly consented customer-support recordings is recommended before production deployment.